

The Fetal Acoustic Environment may be too Noisy for Prenatal Phonetic Learning From External Sources

Ambre Balleroy^{1,2,5}, Ricard Marxer^{3,4,5}, Claire Kabdebon^{2,5} & Thomas Schatz^{1,5}

¹Aix-Marseille Univ, CNRS, LIS, Marseille, France

²Aix-Marseille Univ, CNRS, CRPN, Marseille, France

³Univ Toulon, Toulon, France

⁴ILLS, CNRS, Montréal, Canada

⁵Institute of Language, Communication and the Brain, Aix-Marseille Univ, CNRS, France

Abstract

Prenatal auditory learning is well-documented, but not at the phonetic level. It may be that phonetic information is so degraded in the prenatal acoustic environment that Prenatal Phonetic Learning (PPL) becomes impossible. Yet, this has never been clearly established, and some results suggest directly or indirectly that some degree of PPL may be taking place. In this study, we assess the feasibility of PPL from external sources through extensive computational simulations of early phonetic learning in simulated prenatal and postnatal environments. We examined the learnability of phonetic contrasts in utero, the added value of prenatal exposure for subsequent learning and whether it could explain earlier attunement to vowels than to consonants. Our results provide preliminary evidence against the feasibility of PPL. We discuss alternative hypotheses and the remaining work needed to complement our results.

Keywords: early phonetic learning; prenatal acoustic environment; computational modeling; language acquisition

Introduction

Sensory life begins before birth, when the developing neurons connect to thalamocortical sensory input, approximately on the 24th week of gestation (Kostović & Judoš, 2010). By 28 weeks of gestation, fetuses begin to process both the auditory environment of the womb (Abrams et al., 1998; Parga et al., 2018) and external sounds that filter through (Querleu et al., 1988). They learn from what they perceive in the womb, showing familiarity after birth with stories to which they were exposed and with their mother's voice (DeCasper & Fifer, 1980; DeCasper & Spence, 1986). This prenatal exposure may serve important functions in development. For example, it has been shown to initiate attunement to the prosody of the native language (Abboub et al., 2016).

While prosodic learning is well-documented, it is less clear whether learning is possible prenatally at the phonetic level, since the mother's body acts as a low-pass filter (Gélat et al., 2025; Gerhardt et al., 1990; Querleu et al., 1988) that degrades phonetic information. This is especially the case for consonants, whose acoustic cues are less preserved than those for vowels (Griffiths et al., 1994). Even for vowels, Early Phonetic Learning (EPL), during which infants' perception attunes to the phonetic contrasts of their native language, is typically documented only long after birth, at around 6–9 months of age (Tsuji & Cristia, 2014).

Still, the possibility of Prenatal Phonetic Learning (PPL)

should not be discarded too hastily. First, a study reporting that newborns already respond differently to a vowel from their first language than to a vowel from an unfamiliar language provides direct evidence for it (Moon et al., 2013). In addition, PPL may help account for unexplained phenomena, such as the order in which contrasts are learned. As previously explained, vowels are better preserved in the womb than consonants, and also happen to be learned a few months earlier: around 6–9 months (Tsuji & Cristia, 2014), as opposed to around 10–12 months for consonants (Nazzi & Cutler, 2019; Werker & Tees, 1984), suggesting a possible causal relationship. Finally, PPL might have an adaptive value, with implications for clinical practice. It has been suggested that lack of exposure to a typical Prenatal Acoustic Environment (PAE) in preterm infants may explain some of their language delays (François et al., 2021) and there is evidence that initially degraded input can enhance later perceptual learning in vision (M. Vogelsang et al., 2024) and for prosody (M. Vogelsang et al., 2023). This calls for investigating whether the PAE provides such an *adaptive initial degradation* (L. Vogelsang et al., 2024) of speech for phonetic learning.

Assessing the potential causal role of PPL in subsequent phenomena, such as the delay in learning consonants relative to vowels or language delays in preterm infants is challenging, however, due to the presence of many confounding factors and alternative explanations, which are not easily disentangled in infant experiments. The delayed specialization for consonants, for example, could also be explained by inherent acoustic properties of the phonemes, by frequency effects (Anderson et al., 2003) or by maturational constraints (Menn et al., 2023; Peña et al., 2012). As for the language delays seen in preterms, they may also result from exposure to a lesser amount of speech compared to full-terms (Monson et al., 2023), or from a general sensory deprivation in the neonatal intensive care environment (Jobe, 2014).

Faced with the difficulty, if not impossibility, of controlling for these confounding factors in empirical research, computational modeling provides an attractive alternative. It enables testing the feasibility of theoretical proposals by explicitly controlling each parameter, in order to link them to their long-term consequences, and to identify decisive experiments. While EPL has been successfully simulated using computational modeling (see Matusevych et al. 2023; Poli et al. 2024; Schatz et al. 2021, for example), few modeling works con-

sidered the effect of the PAE, and all but one (Cruz Blandón et al., 2025) used non-realistic input (Chládková et al., 2020; Seebach et al., 1994). None modeled *non-native* perception, whose understanding is necessary for a comprehensive view of first language specialization. Additionally, none considered both vowels and consonants, even though their acquisition timeline may differ (Tsuji & Cristia, 2014). Finally, no existing model includes the intrauterine noise despite its predominance in the PAE (Gerhardt & Abrams, 1996).

In this study, we investigate PPL’s feasibility through explicit learning simulations in the context of *non-native* speech perception of both consonants and vowels. We leverage the empirical literature describing the PAE to create speech training material that simulates what a fetus would hear. We do so by recreating the way outside sounds are filtered by maternal tissue, and mixed with intrauterine noise. We then expose an established model of EPL (Schatz et al., 2021) to various combinations of prenatal- and postnatal-like training material, systematically varying training durations, noise levels and filtering conditions. In Experiment 1, we examine the feasibility of PPL and whether it can explain why perceptual attunement appears to occur later for consonants than for vowels. There is some experimental uncertainty concerning the PAE noise level and, in Experiment 2, we strengthen our results from Experiment 1, by testing the feasibility of PPL under different noise levels. Finally, in Experiment 3, we test if the PAE provides an adaptive initial degradation of external speech signals, facilitating subsequent phonetic learning.

Methods

The current study follows the modeling approach from Schatz et al. (2021). In all cases, we train an unsupervised clustering algorithm on unsegmented speech data either in Japanese or in American English, to simulate learning conditions where an infant is exposed to a given native language. We then quantify the difficulty of discriminating the various contrasts of American English on the basis of the model’s learned representations using a machine ABX task (Schatz, 2016; Schatz et al., 2013). We consider a model to have a native-like perception if it does better at discriminating the contrast from the language it was trained on than the models trained on another language. Our contribution consists of adding the simulation of the PAE to this setup, and evaluating its influence on EPL.

Table 1: Summary of data used for the simulations.

Subset	Dataset	Language	Duration	Number of speakers	Reference
Train	Globalphone	Japanese	19h33	96	Schultz (2002)
	Wall Street Journal	American English	19h30	96	Paul and Baker (1992)
Test	Wall Street Journal	American English	9h39	47	Paul and Baker (1992)

Learning Environment

Training Data We used two datasets of read adult-directed speech from Schatz et al. (2021) as input to our simulations (Table 1). We did not use infant-directed speech, as quality

large-scale datasets are hard to access for other languages than English and since it is unclear whether a learning environment restricted to infant-directed speech is a good model of the PAE. For English, we used the Wall Street Journal (WSJ) corpus (Paul & Baker, 1992), and for Japanese, the Globalphone (GPJ) corpus (Schultz, 2002). The two datasets were split in train and a test subsets, and matched on total duration, speaker number, duration per speaker, and speaker gender (there was no speaker overlap between train and test subsets). We only used short data subsets (at most 1h25), as previous work using the same model (Li et al., 2020) showed this was more than sufficient to see effects emerge, with only marginal improvements in native effects magnitude beyond that, even when trained on more than 100h of data.

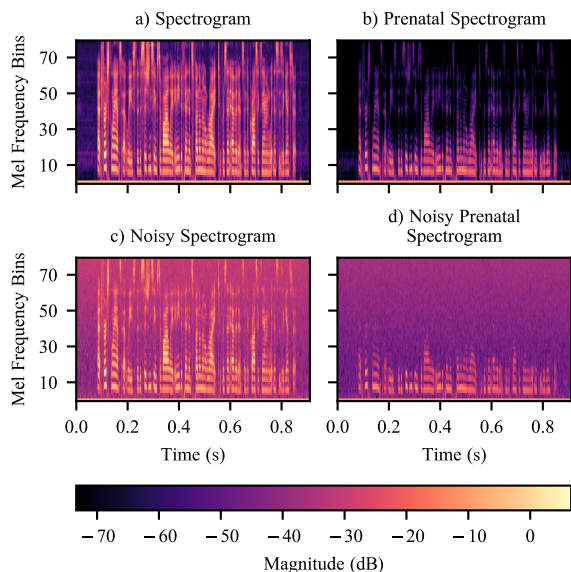


Figure 1: Examples of log-mel spectrograms showcasing our simulation of the PAE for an SNR of 10 dB.

Simulation of the PAE Our goal was to approximate the acoustic environment a fetus is exposed to by simulating the filter created by the maternal tissues, and the noise from the womb environment. Existing studies testing pregnant women or ewes showed the mother’s body acts as a low-pass filter, mainly preserving frequencies below 400-500 Hz (Gélat et al., 2025; Querleu et al., 1988). At higher frequencies, attenuation gradually increases. Other studies (Abrams et al., 1998; Parga et al., 2018) focused on the amount of noise emitted by the mother’s body (cardiac rhythm, breathing...) and hearable in the womb, potentially masking external sounds significantly and affecting learning conditions. We did not try to model the development of fetal audition, as maturational constraints on auditory processing are outside the scope of the current study that focuses on environmental constraints.

To simulate the mother’s body acoustic filter, we derived a transfer function from Figure 5 of Querleu et al. (1988). The

resulting transfer function had a regular slope in log-log units in the high-frequency, which we leveraged to extrapolate the transfer function up to 8,000 Hz (so the frequency range was sufficient to reach the Nyquist frequency of our data sampled at 16,000 Hz). We applied the resulting filter to the training audio files. Examples of raw vs filtered data are displayed in Figure 1 a and b. To design the intrauterine noise, we first estimated the spectral density of intrauterine noise from the Figure 1 of Abrams et al. (1998) in which ewes and fetuses were recorded in silence. We then created colored noise with the spectral shape of intrauterine noise by filtering Gaussian white noise with this spectral profile. This simulated intrauterine noise could subsequently be added to the training data, after applying the mother’s body filter (and not before, as we assume the internal noise is not affected by the external maternal tissues) or independently. We added the noise at various signal-to-noise ratios (SNR) depending on the experiment. Examples of noisy data are displayed in Figure 1c and d.

Learning Mechanism

Raw audio signal with a sampling frequency of 16,000 Hz was chunked in 25-ms-long frames sampled every 10 ms and 13 Mel-Frequency Cepstral Coefficients (MFCCs) were extracted. We concatenated the MFCCs to their 1st and 2nd order derivative, resulting in 39 features. While not traditionally used in linguistics, MFCCs are standards in speech processing, and are a relevant acoustic correlates to vowels (Iskarous & Vietti, 2025; Zahorian & Jagharghi, 1993).

Following previous studies (Matusevych et al., 2023; Schatz et al., 2021) that modeled distributional learning in infants, we applied a Dirichlet Process Gaussian Mixture Model (DPGMM)—an unsupervised clustering algorithm which discovers the number of clusters during training—over single MFCC frames. We initialized the model in the same way as Schatz et al. (2021) and trained for 1,500 iterations as they did, which means the models saw each frame 1,500 times.

Learning outcome assessment

Target phonetic contrasts To answer our question about the different learning timings of native vowel versus consonant contrasts, we needed to choose at least one pair of vowels and one pair of consonants from our test language (American English). Ideally, we would have chosen contrasts that have both been empirically shown as being hard to discriminate for infant learners of Japanese. However, to the best of our knowledge, there exists only one language pair for which non-native empirical discrimination data is available for both vowels and consonants (Bosch & Sebastián-Gallés, 2003a; Bosch & Sebastián-Gallés, 2003b; Singh et al., 2022), and previous computational work failed to simulate the observed vowel effect for this language pair (Matusevych et al., 2023). As a consequence, for consonants, we relied on the well-established [ɹ]-[l] contrast (Kuhl et al., 2006). For vowels, we relied on two American English vowel contrasts that are robustly predicted to be hard to discriminate by Japanese

infants by our model (Schatz et al., 2021): [ɹ]-[ʌ] and [ɹ]-[ɪ] (involving the final r-colored vowel in "mother", and the vowels in "but" and "bid", respectively).

Speech perception task We assessed the discrimination abilities of our ‘native’ (English) and ‘non-native’ (Japanese) models using the machine ABX discrimination task (Schatz, 2016; Schatz et al., 2013). This task measures a discrimination score for a phonetic contrast by forming triplets A, B, X where A is an exemplar of one of the phones, B an exemplar of the other and X is another exemplar of say the same phone as A. If the model’s representation of X is closer to the representation of A than to the representation of B, the model is awarded a score of 1 for this triplet, and a score of 0 otherwise. The overall discrimination score is obtained by averaging these individual triplet scores over a large random sample of all such triplets under the additional constraint that A, B and X always be produced by the same speaker and in the same phonetic context (previous and following sounds).

As in Schatz et al. (2021), a model’s representation of a sound is obtained by extracting MFCC frames from the sound and computing the posterior probability for each learned cluster in the model of having generated each frame. The results are assembled as a sequence of n posterior probability vectors of size d , where n is the number of MFCC frames and d the number of learned clusters. The similarity between model representation is computed using the dynamic time-warping algorithm with a symmetrized Kullback-Leibler divergence.

Statistics For all experiments, we do not display raw ABX scores, but the effect size of native advantage, defined as the effect size (Hedges’ g) of the difference between ABX scores of models trained on Japanese and models trained on American English, and their 95% confidence intervals. Training models under different conditions (e.g. noise-levels) on the same data subsets introduces dependencies that make visual assessment of confidence intervals insufficient to assess whether observed differences are statistically significant. We therefore also perform pairwise permutation tests respecting the structure of the introduced dependencies to evaluate where significant differences are present. For each experiment, we use Benjamini-Yekutieli correction for multiple comparisons to keep the expected false discovery rate under 5%.

Experiment 1

In an initial experiment, we ask whether the PAE is indeed the cause of the delay in learning consonants. To do this, we replicate Schatz et al. (2021), but with the addition of plausible simulations of different components of the PAE: the low-pass filter (Querleu et al., 1988) and the intrauterine noise (Abrams et al., 1998; Parga et al., 2018). We also included a baseline of normal speech to verify that the order in which phonetic contrasts are learnt is not influenced by the inherent acoustic properties of vowels and consonants unrelated to the PAE.

Simulations We began with a prenatal speech condition involving low-pass filtering and the addition of noise. We trained models on four durations, which were evenly spaced on a logarithmic scale ranging from 10 seconds to approximately 1h25. For each duration, 8 different models were trained using non-overlapping subsets of the training data, except for the models trained on 1h25, for which we could only obtain 5 non-overlapping subsets. We then ran simulations using the same procedure but with postnatal (unfiltered) speech, postnatal speech with added noise and prenatal speech without added noise as a baseline. Since these Prenatal-only or Postnatal-only conditions are not representative of the learning experience of infants, who are exposed to both conditions sequentially, we also trained models on 1h25 of Prenatal data followed by various duration of Postnatal data (Prenatal + Postnatal condition), with no overlap between the prenatal and postnatal data.

For our simulations to be plausible, we needed to calculate an SNR value for intrauterine noise based on empirical evidence. We did this by comparing the difference in intensity between the filtered speech signal and the intrauterine background noise. To simulate the filtered speech spectrum, we extracted the long-term average speech spectra from Byrne et al. (1994) and applied the body filter data from Querleu et al. (1988) to it. We then selected one existing description of the intrauterine noise frequency spectrum: the 'Lying' condition in Figure 2 of Parga et al. (2018). From this, we integrated over the available frequency range to obtain overall intrauterine SNRs of 11.8 and 11.2 dB, respectively, for male and female voices. Consequently, we used an SNR of 10 dB in our noised simulations.

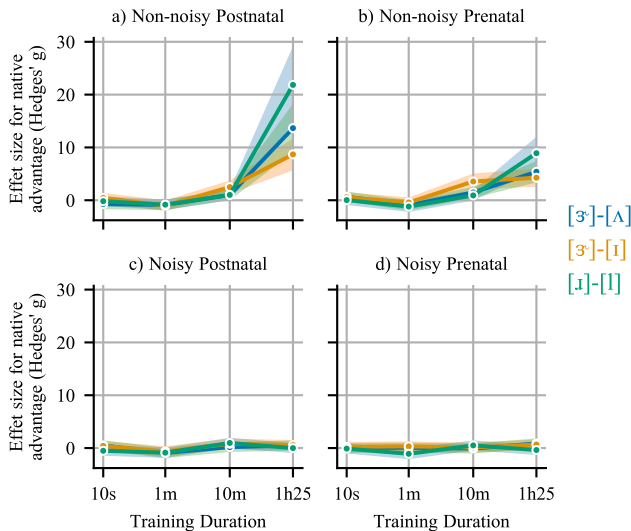


Figure 2: Effect size of native advantage as a function of training duration, added noise, contrast, and condition (prenatal or postnatal). The SNR used for adding noise is 10 dB. Error bars: 95% confidence intervals.

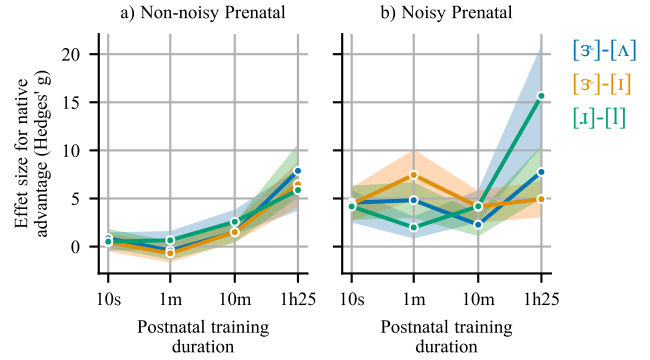


Figure 3: Effect size of native advantage for models trained first on 1h25 of prenatal speech, as a function of postnatal training duration, added noise, and contrast. The SNR used for adding noise is 10 dB. Error bars: 95% confidence intervals.

Results We first needed to verify in which condition a reliable native advantage could emerge. We started by considering the Prenatal-only and Postnatal-only conditions. We asked if, for a given SNR, contrast, and condition, the effect size of the native advantage was larger for longer training duration. This was not the case in any noisy simulations, where no comparison proved significant (Figure 2c and d). In the non-noisy conditions, some comparisons proved significant (see Table 2 and Figure 2a and b). They showed the models trained on 1h25 always learned a significantly larger native advantage than models trained on 10 seconds, and, in most cases, on 1 minute. Native advantages for models trained on 10 minutes of data were also significantly larger than those for models trained on 1 minute in all conditions except one. Overall, this means the models trained on larger durations learned a reliable native effect in non-noisy conditions, but it was less the case for noisy conditions, regardless of the low-pass filter.

We also tested for significant differences across contrasts for a given training duration, SNR and condition, to test whether the native advantage for vowels emerged earlier than the advantage for consonants. No significant effect was found.

Finally, we verified there was still no earlier emergence of vowels in the most realistic simulation, the Prenatal + Postnatal condition (Figure 3). We tested for significant differences across contrasts for a given postnatal training duration and prenatal SNR. Again, no significant differences were found.

These negative results suggest that the empirically observed delay for consonants is due neither to intrinsic properties of vowels that would make them easier to learn, nor to an interaction with prenatal learning, since they were not learned earlier than consonants in any conditions. The absence of any prenatal learning in realistic (noisy) conditions led us to question our choice of SNR value. As a consequence, we report simulations for different SNRs in the next section, to investigate whether a higher SNR could enable us to match existing evidence of prenatal EPL (Moon et al., 2013).

Table 2: Significant differences found in Experiment 1 after correction for multiple comparisons for the non-noisy models, when comparing the effect sizes of models trained on two training duration for a given contrast and condition.

Condition	Contrast	Compared Durations	Mean Effect Size Difference	P-value
Prenatal	[ʒ]-[i]	10s - 1m	3.11	<0.001
		10s - 1h25	4.81	<0.001
		1m - 10m	3.49	<0.002
		1m - 1h25	5.19	<0.001
	[ʒ]-[A]	10s - 1h25	3.35	<0.002
		1m - 1h25	4.38	<0.002
Postnatal	[i]-[l]	10s - 1h25	4.05	<0.002
		1m - 1h25	5.02	<0.001
	[ʒ]-[i]	10s - 1h25	6.06	<0.001
		1m - 10m	3.23	<0.001
		1m - 1h25	7.23	<0.001
	[ʒ]-[A]	10s - 1h25	7.98	<0.001
		1m - 1h25	7.53	<0.001
	[i]-[l]	10s - 1h25	9.32	<0.001
		1m - 1h25	9.63	<0.001
		10m - 1h25	8.02	<0.001

Experiment 2

In the previous section, we replicated Schatz et al. (2021) using a plausible simulation of the PAE, but no prenatal learning was observed in realistic conditions. In the second set of simulations, we test different SNRs in order to find which one best matches the empirical records (Moon et al., 2013).

Simulations For this set of simulations, we used 7 training data subsets, each 1h25 long. We varied the SNR from 0 to 40 dB in order to find a sufficient one for phonetic learning to occur. We also trained models on non-noisy conditions, which correspond to 60 dB, the approximate intensity of a normal conversation in a quiet environment.

Results We tested for significant differences across SNR values and within contrasts and condition (Prenatal and Postnatal). All significant comparisons are summarized in Table 3, and native advantages displayed in Figure 4. In most cases, the 60 dB scores were significantly better than the 10 dB scores, and at least one significant pair showed a greater native advantage when the SNR value increased. For the Postnatal models, there were no significant differences between the 30, 40, and 60 dB scores. Consequently, a reliable native advantage could be demonstrated in Postnatal conditions with a low SNR of 30 dB. In Prenatal conditions, at least one SNR comparison proved significant from 60 dB for each contrast, which also shows that a larger native advantage was learned when increasing the SNR, albeit in a less conclusive way. While postnatal models developed a stable native advantage with an SNR value as low as 30 dB, this was less the case for prenatal models, for which the effects were less reliable, especially at lower SNR values. In other words, the filter, in the prenatal conditions, made the models more sensitive to noise.

We also tested for differences between contrasts within SNR values and conditions, with no significant results, and thus no evidence that vowels are easier to learn than consonants.

In this experiment, learning any native advantage in plausible conditions proved challenging (as an SNR of 60 dB is unplausible in the PAE). However, exposure to prenatal-like speech, while not immediately beneficial, may still influence postnatal EPL. We examine this possibility in the next section.

Table 3: Significant differences found in Experiment 2 after correction for multiple comparisons, when comparing the effect sizes of models trained on different SNRs, for a given contrast and condition.

Condition	Contrast	Compared SNRs	Mean Effect Size Difference	P-value
Prenatal	[ʒ]-[i]	0 dB - 60 dB	7.28	<0.001
		10 dB - 60 dB	5.71	<0.002
		20 dB - 60 dB	6.41	<0.002
		30 dB - 60 dB	5.44	<0.003
	[ʒ]-[A]	10 dB - 60 dB	3.49	<0.003
Postnatal	[i]-[l]	0 dB - 60 dB	4.68	<0.002
		20 dB - 60 dB	4.54	<0.001
	[ʒ]-[i]	0 dB - 30 dB	8.71	<0.001
		0 dB - 40 dB	8.00	<0.001
		0 dB - 60 dB	6.73	<0.002
	[ʒ]-[A]	0 dB - 30 dB	5.36	<0.003
		0 dB - 40 dB	6.89	<0.001
		0 dB - 60 dB	7.29	<0.003
	[i]-[l]	0 dB - 30 dB	6.06	<0.003
		0 dB - 40 dB	5.37	<0.002
		10 dB - 30 dB	5.41	<0.003

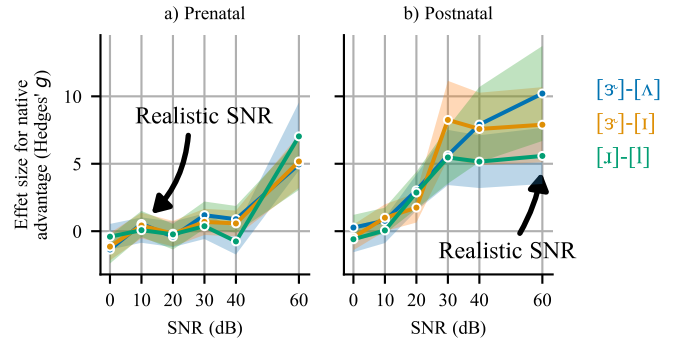


Figure 4: Effect size of native advantage as a function of SNR value in Postnatal and Prenatal condition for the target contrasts. Error bars: 95% confidence intervals.

Experiment 3

In the second experiment, models trained on prenatal conditions struggled to learn any native advantage unless they were not exposed to any noise, which is unrealistic. Despite the apparent difficulty of learning in the prenatal environment, it is possible that exposure to a PAE may be beneficial to infants in the long term after birth. In this section, we run simulations where models are trained on prenatal speech followed by postnatal speech, in order to model the effect of birth. If exposure to the PAE is also beneficial for EPL, we should see a larger native advantage in models trained on a longer duration of prenatal speech when the postnatal duration is kept constant.

This comes in contrast to Experiment 1, where we varied the postnatal duration but kept prenatal duration constant.

Simulations For this experiment, we used concatenations of prenatal (filtered and noisy using the SNRs from Experiment 1) and postnatal (unfiltered and non-noisy) speech. For prenatal exposure, we used the same durations and subsets number as in Experiment 1. For the postnatal exposure, we used the longest duration at our disposal (about 1h25).

Results We tested the native advantages for significant differences within contrasts and across prenatal pretraining duration. No significant effect was found, suggesting the effect size of native advantage does not depend on the amount of prenatal training (Figure 5). We thus find no evidence that prenatal exposure offers any advantage for postnatal EPL¹.

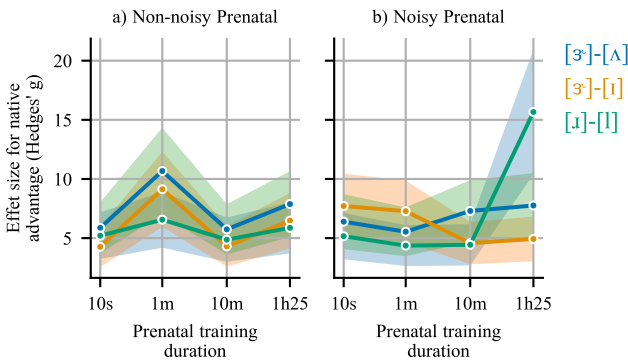


Figure 5: Effect sizes of native advantage as a function of prenatal training duration for models trained on 1h25 of postnatal data in noisy (10 dB) and non-noisy (60 dB) conditions. Error bars: 95% confidence intervals.

Discussion

Our simulations of the effect of the PAE on EPL showed no condition in which the native advantage for vowels appeared before that for consonants. If our results hold for human infants, this suggests the delayed specialization for consonants observed empirically (Tsuji & Cristia, 2014) may not be the consequence of prenatal speech exposure, but of other factors, such as maturational constraints (Menn et al., 2023). Alternatively, our model could be incorrect, failing for vowels, despite its success at modeling native specialization for consonants (Matusevych et al., 2023; Schatz et al., 2021). Indeed, the model predicts an effect for the vowel contrasts considered, but this has not been tested empirically. Moreover, we are not aware of any successful computational modeling for documented vowel effects (Matusevych et al., 2023). Findings of a mechanism differentially predictive for consonant and vowel effects would be remarkable in itself and this deserves

further investigation, through speech perception experiments assessing the empirical validity of the model's predictions on vowel contrasts and/or through simulations assessing if the model correctly predicts documented vowel effects for other languages. Comparisons with alternative implementations of EPL that could result in different outcomes and are equally credible (Cruz Blandón et al., 2025; Lavechin et al., 2025; Matusevych et al., 2023), would also be informative.

This study also modeled the feasibility of PPL under plausible noise condition. We showed that even with better SNR than considered realistic in the literature, outside voices are too degraded for the fetus to learn phonetic information from them. We also found no benefits of prenatal exposure for postnatal EPL. Since a single study in the literature shows PPL (Moon et al., 2013), it remains possible for it to be a false positive, and replication attempts would be helpful. As for the delays in phonetic learning observed in preterms they may be caused by other factors than the lack of exposure to the PAE, in particular the sensory deprivation they often experience (Jobe, 2014; Monson et al., 2023). If these interpretations hold, the phonetic domain would contrast with other domains in perceptual learning and language acquisition, where initial input degradation or simplification is adaptive (Elman, 1993; Newport, 1988; M. Vogelsang et al., 2023).

There are alternative interpretations for our results that may be compatible with Moon et al. (2013) and an adaptive role for PPL, however. First, we used colored noise, which does not take into account the amplitude modulations of real intrauterine noise. These could provide regular dips in the noise level during which cleaner signal would be able to reach the fetus and enable PPL. Second, even if external voices do not support PPL, maternal voice might (Kepinska, 2025). Indeed, unlike external voices, it is amplified in utero (Richards et al., 1992), making learning based on it more plausible. This comes in contrast to the most discussed scenario for prenatal language acquisition, according to which all input is low-pass filtered (Gervain, 2018). Finally, different modeling choices might yet lead to different outcomes: While infants' perception may adapt to the sudden change in input acoustical properties at birth to facilitate continual learning, we did not implement adaptation mechanisms; The temporal window over which the learning mechanism operates might matter, but we considered only 10ms-long MFCC frames with 1st and 2nd time derivatives; We tested DPGMMs, but neural network mechanisms have also been proposed (Lavechin et al., 2025; We did not model fetus' bone conduction hearing, which would lead to even stricter high frequencies filtering (Gerhardt & Abrams, 1996). Follow-up work will explore these possibilities through additional simulations.

In summary, this study provides preliminary evidence against the feasibility of PPL and will serve as important groundwork towards exploring refined hypotheses regarding the potential roles of the mother's voice and of intrauterine noise amplitude modulations in PPL, as well as regarding the validity of DPGMMs as models of EPL.

¹Native ABX discrimination scores also showed no benefit of prenatal exposure. Results not shown for lack of space.

Acknowledgements

This work, carried out within the Institute of Convergence ILCB (ANR-16-CONV-0002), has benefited from support from the French government (France 2030), managed by the French National Agency for Research (ANR) and the Excellence Initiative of Aix-Marseille University (A*MIDEX). Additionally, it benefited from the support of the French National Research Agency through the ANR-20-CE230012-01 (MIM) and ANR-25-CE28-4388-01 (GESPER) grants. It also received support from the French government under the France 2030 investment plan, as part of the Initiative d'Excellence d'Aix-Marseille Université – A*MIDEX AMX-21-PEP-021 (DEV-LEARN). We would like to thank Olga Kepinska for pointing out the importance of the maternal voice in the PAE and for helpful discussion. We acknowledge the use of ChatGPT as an occasional coding completion and troubleshooting assistant.

References

- Abboub, N., Nazzi, T., & Gervain, J. (2016). Prosodic grouping at birth. *Brain and Language*, 162, 46–59. <https://doi.org/10.1016/j.bandl.2016.08.002>
- Abrams, R. M., Gerhardt, K. J., Griffiths, S. K., Huang, X., & Antonelli, P. J. (1998). Intrauterine Sounds in Sheep. *Journal of Sound and Vibration*, 216(3), 539–542. <https://doi.org/10.1006/jsvi.1998.1755>
- Anderson, J. L., Morgan, J. L., & White, K. S. (2003). A Statistical Basis for Speech Sound Discrimination. *Language and Speech*, 46(2–3), 155–182. <https://doi.org/10.1177/00238309030460020601>
- Bosch, L., & Sebastián-Gallés, N. (2003a). Simultaneous Bilingualism and the Perception of a Language-Specific Vowel Contrast in the First Year of Life. *Language and Speech*, 46(2–3), 217–243. <https://doi.org/10.1177/00238309030460020801>
- Bosch, L., & Sebastián-Gallés, N. (2003b). Language experience and the perception of a voicing contrast in fricatives: Infant and adult data. *Proceedings of the 15th International Conference of Phonetic Sciences*, 1987–1990. Retrieved May 17, 2024, from https://www.internationalphoneticassociation.org/icphs-proceedings/ICPhS2003/papers/p15_1987.pdf
- Byrne, D., Dillon, H., Tran, K., Arlinger, S., Wilbraham, K., Cox, R., Hagerman, B., Hetu, R., Kei, J., Lui, C., Kiessling, J., Kotby, M. N., Nasser, N. H. A., El Kholy, W. A. H., Nakanishi, Y., Oyer, H., Powell, R., Stephens, D., Meredith, R., . . . Ludvigsen, C. (1994). An international comparison of long-term average speech spectra. *The Journal of the Acoustical Society of America*, 96(4), 2108–2120. <https://doi.org/10.1121/1.410152>
- Chládková, K., Nudga, N., & Boersma, P. (2020). A Model of Prenatal Acquisition of Vowels. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 42.
- Cruz Blandón, M. A., Gonzalez-Gomez, N., Lavechin, M., & Räsänen, O. (2025). Simulating prenatal language exposure in computational models: An exploration study. *Cognition*, 256, 106044. <https://doi.org/10.1016/j.cognition.2024.106044>
- DeCasper, A. J., & Fifer, W. P. (1980). Of Human Bonding: Newborns Prefer Their Mothers' Voices. *Science*, 208(4448), 1174–1176. <https://doi.org/10.1126/science.7375928>
- DeCasper, A. J., & Spence, M. J. (1986). Prenatal maternal speech influences newborns' perception of speech sounds. *Infant Behavior and Development*, 9(2), 133–150. [https://doi.org/10.1016/0163-6383\(86\)90025-1](https://doi.org/10.1016/0163-6383(86)90025-1)
- Elman, J. L. (1993). Learning and development in neural networks: The importance of starting small. *Cognition*, 48(1), 71–99. [https://doi.org/10.1016/0010-0277\(93\)90058-4](https://doi.org/10.1016/0010-0277(93)90058-4)
- François, C., Rodriguez-Fornells, A., Teixidó, M., Agut, T., & Bosch, L. (2021). Attenuated brain responses to speech sounds in moderate preterm infants at term age. *Developmental Science*, 24(1), e12990. <https://doi.org/10.1111/desc.12990>
- Gélat, P., van't Wout, E., Haqshenas, R., Melbourne, A., David, A. L., Mufti, N., Henriques, J., Thibaut de Maisières, A., & Jauniaux, E. (2025). Evaluation of fetal exposure to environmental noise using a computer-generated model. *Nature Communications*, 16(1), 3916. <https://doi.org/10.1038/s41467-025-58983-0>
- Gerhardt, K. J., & Abrams, R. M. (1996). Fetal hearing: Characterization of the stimulus and response. *Seminars in Perinatology*, 20(1), 11–20. [https://doi.org/10.1016/S0146-0005\(96\)80053-X](https://doi.org/10.1016/S0146-0005(96)80053-X)
- Gerhardt, K. J., Abrams, R. M., & Oliver, C. C. (1990). Sound environment of the fetal sheep. *American Journal of Obstetrics and Gynecology*, 162(1), 282–287. [https://doi.org/10.1016/0002-9378\(90\)90866-6](https://doi.org/10.1016/0002-9378(90)90866-6)
- Gervain, J. (2018). The role of prenatal experience in language development. *Current Opinion in Behavioral Sciences*, 21, 62–67. <https://doi.org/10.1016/j.cobeha.2018.02.004>
- Griffiths, S. K., Brown, W. S., Gerhardt, K. J., Abrams, R. M., & Morris, R. J. (1994). The perception of speech sounds recorded within the uterus of a pregnant sheep. *The Journal of the Acoustical Society of America*, 96(4), 2055–2063. <https://doi.org/10.1121/1.410147>
- Iskarous, K., & Vietti, A. (2025). Phonetic information in the vowel spectrum: The meaning of Mel-Frequency Cepstral Coefficients. *Journal of Phonetics*, 112, 101434. <https://doi.org/10.1016/j.wocn.2025.101434>
- Jobe, A. H. (2014). A Risk of Sensory Deprivation in the Neonatal Intensive Care Unit. *The Journal of Pediatrics*, 164(6), 1265–1267. <https://doi.org/10.1016/j.jpeds.2014.01.072>
- Kepinska, O. (2025). *Diverse Sounds: How Prenatal Language Experience Shapes Fetal Brain Development and Later Language Learning. A Longitudinal Study.* (Grant

- Agreement No. 101221485) [Unpublished ERC Grant Agreement]. <https://doi.org/10.3030/101221485>
- Kostović, I., & Judaš, M. (2010). The development of the subplate and thalamocortical connections in the human foetal brain. *Acta Paediatrica*, 99(8), 1119–1127. <https://doi.org/10.1111/j.1651-2227.2010.01811.x>
- Kuhl, P. K., Stevens, E., Hayashi, A., Deguchi, T., Kiritani, S., & Iverson, P. (2006). Infants show a facilitation effect for native language phonetic perception between 6 and 12 months. *Developmental Science*, 9(2), F13–F21. <https://doi.org/10.1111/j.1467-7687.2006.00468.x>
- Lavechin, M., De Seyssel, M., Titeux, H., Wisniewski, G., Bredin, H., Cristia, A., & Dupoux, E. (2025). Simulating early phonetic and word learning without linguistic categories. *Developmental Science*, 28(2), e13606.
- Li, R., Schatz, T., Matushevych, Y., Goldwater, S., & Feldman, N. H. (2020). Input matters in the modeling of early phonetic learning. *Proceedings of the Annual Conference of the Cognitive Science Society*.
- Matushevych, Y., Schatz, T., Kamper, H., Feldman, N. H., & Goldwater, S. (2023). Infant Phonetic Learning as Perceptual Space Learning: A Crosslinguistic Evaluation of Computational Models. *Cognitive Science*, 47(7), e13314. <https://doi.org/10.1111/cogs.13314>
- Menn, K. H., Männel, C., & Meyer, L. (2023). Does Electrophysiological Maturation Shape Language Acquisition? *Perspectives on Psychological Science*, 18(6), 1271–1281. <https://doi.org/10.1177/17456916231151584>
- Monson, B. B., Ambrose, S. E., Gaede, C., & Rollo, D. (2023). Language Exposure for Preterm Infants is Reduced Relative to Fetuses. *The Journal of Pediatrics*, 262, 113344. <https://doi.org/10.1016/j.jpeds.2022.12.042>
- Moon, C., Lagercrantz, H., & Kuhl, P. K. (2013). Language experienced in utero affects vowel perception after birth: A two-country study. *Acta Paediatrica*, 102(2), 156–160. <https://doi.org/10.1111/apa.12098>
- Nazzi, T., & Cutler, A. (2019). How Consonants and Vowels Shape Spoken-Language Recognition. *Annual Review of Linguistics*, 5(1), 25–47. <https://doi.org/10.1146/annurev-linguistics-011718-011919>
- Newport, E. L. (1988). Constraints on learning and their role in language acquisition: Studies of the acquisition of American sign language. *Language Sciences*, 10(1), 147–172. [https://doi.org/10.1016/0388-0001\(88\)90010-1](https://doi.org/10.1016/0388-0001(88)90010-1)
- Parga, J. J., Daland, R., Kesavan, K., Macey, P. M., Zeltzer, L., & Harper, R. M. (2018). A description of externally recorded womb sounds in human subjects during gestation (J. P. Van Wouwe, Ed.). *PLOS ONE*, 13(5), e0197045. <https://doi.org/10.1371/journal.pone.0197045>
- Paul, D. B., & Baker, J. M. (1992–February 26). The Design for the Wall Street Journal-based CSR Corpus. *Speech and Natural Language: Proceedings of a Workshop Held at*, 6, 1992.
- Peña, M., Werker, J. F., & Dehaene-Lambertz, G. (2012). Earlier Speech Exposure Does Not Accelerate Speech Acquisition. *The Journal of Neuroscience*, 32(33), 11159–11163. <https://doi.org/10.1523/JNEUROSCI.6516-11.2012>
- Poli, M., Schatz, T., Dupoux, E., & Lavechin, M. (2024). Modeling the initial state of early phonetic learning in infants. *Language Development Research: An Open-Science Journal*, 4(2). <https://doi.org/10.34842/Y89T-6Q31>
- Querleu, D., Renard, X., Versyp, F., Paris-Delrue, L., & Crèpin, G. (1988). Fetal hearing. *European Journal of Obstetrics & Gynecology and Reproductive Biology*, 28(3), 191–212. [https://doi.org/10.1016/0028-2243\(88\)90030-5](https://doi.org/10.1016/0028-2243(88)90030-5)
- Richards, D. S., Frentzen, B., Gerhard, K. J., McCANN, M. E., & Abrams, R. M. (1992). Sound Levels in the Human Uterus. *Obstetrics & Gynecology*, 80(2), 186. Retrieved July 8, 2025, from https://journals.lww.com/greenjournal/abstract/1992/08000/Sound_Levels_in_the_Human_Uterus.6.aspx
- Schatz, T. (2016, September 29). *ABX-discriminability measures and applications* [These de doctorat]. Paris 6. Retrieved April 28, 2026, from <https://theses.fr/2016PA066324>
- Schatz, T., Feldman, N. H., Goldwater, S., Cao, X.-N., & Dupoux, E. (2021). Early phonetic learning without phonetic categories: Insights from large-scale simulations on realistic input. *Proceedings of the National Academy of Sciences*, 118(7), e2001844118. <https://doi.org/10.1073/pnas.2001844118>
- Schatz, T., Peddinti, V., Bach, F., Jansen, A., Hermansky, H., & Dupoux, E. (2013). Evaluating speech features with the minimal-pair ABX task: Analysis of the classical MFC/PLP pipeline. *Interspeech 2013*, 1781–1785. <https://doi.org/10.21437/Interspeech.2013-441>
- Schultz, T. (2002). Globalphone: A multilingual speech and text database developed at karlsruhe university. *7th International Conference on Spoken Language Processing (ICSLP 2002)*, 345–348. <https://doi.org/10.21437/ICSLP.2002-151>
- Seebach, B. S., Intrator, N., Lieberman, P., & Cooper, L. N. (1994). A model of prenatal acquisition of speech parameters. *Proceedings of the National Academy of Sciences*, 91(16), 7473–7476. <https://doi.org/10.1073/pnas.91.16.7473>
- Singh, L., Rajendra, S. J., & Mazuka, R. (2022). Diversity and representation in studies of infant perceptual narrowing [Supplemental Material]. *Child Development Perspectives*, 16(4), 191–199. <https://doi.org/10.1111/cdep.12468>
- Tsuji, S., & Cristia, A. (2014). Perceptual attunement in vowels: A meta-analysis. *Developmental Psychobiology*, 56(2), 179–191. <https://doi.org/10.1002/dev.21179>
- Vogelsang, L., Vogelsang, M., Pipa, G., Diamond, S., & Sinha, P. (2024). Butterfly effects in perceptual development: A review of the ‘adaptive initial degradation’ hypothesis. *Developmental Review*, 71, 101117. <https://doi.org/10.1016/j.dr.2024.101117>
- Vogelsang, M., Vogelsang, L., Diamond, S., & Sinha, P. (2023). Prenatal auditory experience and its sequelae. *De-*

- developmental Science*, 26(1), e13278. <https://doi.org/10.1111/desc.13278>
- Vogelsang, M., Vogelsang, L., Gupta, P., Gandhi, T. K., Shah, P., Swami, P., Gilad-Gutnick, S., Ben-Ami, S., Diamond, S., Ganesh, S., & Sinha, P. (2024). Impact of early visual experience on later usage of color cues. *Science (New York, N.Y.)*, 384(6698), 907–912. <https://doi.org/10.1126/science.adk9587>
- Werker, J. F., & Tees, C. (1984). Cross-Language Speech Perception: Evidence for Perceptual Reorganization During the First Year of Life. *Infant Behavior and Development*, 7, 49–63.
- Zahorian, S. A., & Jagharghi, A. J. (1993). Spectral-shape features versus formants as acoustic correlates for vowels. *The Journal of the Acoustical Society of America*, 94(4), 1966–1982. <https://doi.org/10.1121/1.407520>